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Okay, guys, good morning, and thank you for joining me in Week 11. And I hope you can hear me fine. There's no problem with the audio or anything. Just in case you have any issue, just drop me a message on the chat, okay? I'll try to fix things at this end. So welcome to Week 11 content. As you know, this is our last week that we have a lecture. So today, instead of reviewing all previous content, I decided to have a new topic for you. The reason is that you already submitted most of your tasks and already mostly the unhedged tasks have remained. And those tasks do not have anything to do with today's content, okay? This is for your own journal knowledge and to know a bit more of modern topics in AI, specifically human-centered AI, you do not need this knowledge to complete your tasks, even HD1. Okay? Just clarify this first. And besides that, I believe instead of reviewing all previous repetitive topics, it's good to have something new, even if it's last session. So you may carry the knowledge that you I earned today for your more advanced topics during your course, and I hope it helps you understand AI area more profoundly. So I can see we have enough people here. So let's start. As I said, human-centered and artificial intelligence are the main focus of today's topic. So the agenda includes what human-centered AI is, and we talk about a bit of ethical and social harm that it has caused already. Then we move towards responsible AI, which is one of the modern topics we have in AI domain. I also talk about AI agents. That is a very hot topic these days in AI. Then there is a delicate difference between AI model and AI agent. I briefly touched this topic as well. And yeah, I also have a small activity for you to work on later.

02:37

So let's talk about human-centered AI. This human-centered AI topic simply means all AI tools should be developed around human value system. You know that as human, we have many factors and values that probably machine, not probably, definitely machines are unable to understand. For instance, privacy, freedom, safety, dignity, well-being, sense of belonging. All these things should be respected by AI. But this is up to us to teach AI how to do that. Without that, AI can make very bad decisions. I'm going to show you some examples of what can go wrong when we are using AI. You know that technically AI can be very helpful, but it's possible that if there is not enough human intervention, AI make very bad and harmful decisions.

03:37

For instance, this is a real case that happened in USA, okay? There was a hurricane, and people were desperate to escape the location of, you know, hurricane. But you know what happened? There was one built-in system that regulates ticket price. And you know that the regular ticket was like 547, but the automated and AI built-in system decreased the price up to and more than \$3,000. This was something that AI decided. And you see how unfair it can be. Because people are desperate to leave, you know, this zone. But price goes up crazily. So it's very unethical. If it was a human, probably they never made this decision. Right? I just want to let you know, even if AI works well, it doesn't mean that it works ethically, okay? So it made very unfair decision. This was only one of the examples that AI decided. But even we have a good system, what people behind that can decide some evil decisions?

04:57

For instance, Volkswagen had a terrible scandal in 2015. You know what happened? The system was working fine. Just let me see if you can hear me well or not. Okay, I hope things are fine here. Good. These people have designed the system deliberately deceptive, which means they planned AI system in their car to work differently in lab and difference in real-world scenarios while people are driving their cars, to have different level of authority or control on the car. And this was recognized by experts, okay? Then after this scandal, they had to face severe consequences. You know, there is a terrible drop in the rate of selling their cars. The CEO had to resign, and they had to pay a huge amount of financial fine, and they lost reputation. That's the most important thing. So AI is a tool that can be used by wrong people. Okay? So not only AI should work well itself, it also should be used by right, sensible, and ethical people. So there are many layers that we should use when we are trusting AI.

06:49

Another example that was very common recently was an app during COVID. You know, during that time, we had severe quarantine. People were not allowed to leave house unless for very reasonable, you know, purposes. Back then, some apps came to the picture like this COVIDSafe app. And, you know, this app was at the end recognized it was collecting some information from users that they did not need it at all. So they were collecting data without any clear purpose. And you know that these data can

be used for wrong purposes. So this is another concern that we should consider. You shouldn't trust all apps blindly. And when you are importing any data or revealing any details, you should ensure why it's needed in the first place. OK, as a person who is educated in this area from now on, you should be more careful about these aspects of AI. So things can go seriously wrong.

08:04

For instance, these team are some executives in TikTok and they were called in court of justice because, you know, what content was provided in TikTok was leading to exploiting children. You know, we have recommendation system built in most AI systems. And, you know, this recommendation system can have very negative impact. Let's assume that a person has a very bad day, and they got some suicidal thoughts, okay? They are thinking very wrong, you know, very wrong about they should end their lives or something like this. If they search for something like this, recommendation system keeps sending them same wrong, you know, videos or messages or anything. So this can improve this intention in them. So these recommendation system can work evil. And yeah, they can reinforce some bad decisions in people. So these executives were called in the court to justify why they are using this. So you see it can be very serious.

09:15

Another wrong decision that AI can make, and it's very, very severe, is something that is built in some criminal risk system, okay, in USA government. US police is doing this, you know. They wrongly label black people with higher risk. If you compare these people, you see, this white person has low risk. Let's say we rank from 1 to 10. He gets 3, which is less than 5. So we consider low risk. But look, he had drug possessions. He had burglary. Okay? But look at these women. These women didn't have done anything wrong previously, but still she got eight. So you see, she has less severe offenses, but still she got much higher rank comparing this guy. This is something that probably related to some historical data. You know that when you train your AI model, you use historical data, which means data that has been collected previously. And maybe previously, black people in the USA had more crime rates. So model has been trained to label these people high risk. So this is very unfair. OK, this is something that I should be aware of.

10:53

Another example is models are usually trained, especially facial recognition one, because, you know, they are fed by what we gave them. They are mainly trained by male pictures, okay? And when it comes to male, there is also another layer of difference between white and black. So the most data that they got is from white male. So they can do a better job when it comes to this category. And they do the worst job for women and especially for darker skin women. So all these things raise very serious issues when it comes to AI. So facial recognition system can not be that trustworthy either. And it's not always that serious. You know, we talked about TikTok exiting that they called to court. But not all of these issues can make a headline. Sometimes we have some very minor issues that quietly reduce dignity, inclusion, and sense of belonging. For instance, gender equality. Sometimes, let's talk about our university. Maybe previously they didn't ask about gender. You want to address as he, she, or otherwise. If a system doesn't have this, people have not been included, we feel excluded. OK, the sense of belonging can disappear over time. So despite not being a very hot topic in media or headline or in the court or news, still it can impact. So there are many other issues like this that we never discussed, but they are impacting people.

12:50

So all these leads us to have a hot topic, which is called human-centered AI, which means all AI should develop around human needs, abilities, and values. As you noticed, AI can, you know, open very new probability or, you know, opportunity for us, but still it can open very dark possibilities as well, as you noticed. So the purpose of human-centered AI is empowering people and give them sense of support and extend their ability. AI is not there to limit people. It's there to empower us, okay? That's the main purpose of having human-centered AI. So there are different layers or hierarchy system here. How we can have a good human-centered AI? The first layer is about the people who design this. For instance, good engineering. Those people who are actually developing tools, they should consider this. They should test the models and monitor them well to ensure that they have fair data. Let's say facial recognition one, they should have enough or let's say balanced number of pictures from different groups, different race, different gender to train model unbiased. Okay? The model should be fair. If you have only male pictures, the model would be biased towards men.

14:36

The next layer is about the culture of organizations. So engineers have designed the AI. Next layer is organization, that they should consider the safety and learn from the failures that they previously had. So they should have some, let's say, strategies and regulations to handle things to ensure that what they are producing is trustworthy and reliable. The next layer would be external companies that they work independently. They will observe and oversight what has been produced by the other company. If they are buying a product from AI company, they should have their own LERA filtering to see if it's performing well or not. They cannot have, you know, AI tools and use it and they say, oh, I didn't know that this AI is not trustworthy. All people here are responsive and accountable. They should take the responsibility of what they are buying or using. Okay? They cannot leave it to the algorithm. They cannot say, oh, algorithm was wrong. No. It was your responsibility to be aware of what you are getting yourself into. And at the very top level, government also should have some rules for using different AI tools. So everyone here is involved and have their own responsibility and accountability.

16:22

So considering human-centered AI, I want you decision-making process, or last week we talked about predictive AI and generative AI. Both of them actually should be considered here as human and AI both can take role. It's not a one-dimensional model. I cannot say, okay, AI decided that, so I don't have a say. Okay? So this is not a one-dimensional model. For a very complex situation, we never can say, okay, let's move towards computer automation because it can do things much quicker and faster and probably more efficient. It's not trustworthy because what if something goes wrong? Human also should have a level of authority here. So it's not like deciding between human or computer. This is not ideal system. Previously were thinking, okay, if we have automation, that means human doesn't have a say. Let's think of the example that we had last week, automated door. They can open or close. But what if there is a burglary and someone is rushing out of the building? If door is fully automated, okay, it's probably or I think to a great probability, it lets, you know, theft to escape easily. But if it was a human involved, it definitely didn't happen. So it's not good enough to have a one-layer or one-dimension system.

18:29

The best approach is actually having a two-dimensional system. In two-dimensional,

as you see, we have both human control and computer automation, which means we don't have to choose between automation and human control. Both of them should be included at the same time. But how? You know, this quadrant top right is the most preferable quadrant that we need to consider for having complex systems. Which means at the same time that we have highly automated system, which means AI can take care of those hard calculation and being efficient, it's there, but still we have high level of human control. If anything goes wrong, AI should call for help. It needs human assistance. Okay? So that's the best part that we need for very complex works. But there are some works, let's say, something that is highly predictable and straightforward. Let's say doing routine calculation. In here, we can have very high level of computer automation and very low human control. If you're going to use calculator, you trust your calculator easily, right? There's not much of human intervention there. So it depends on the problem that you are facing, okay? And gradually you will learn what level of control is needed by human depending on how well established a problem is. Maybe a problem has been solved many times and you can trust computer that it can take care of it. In that cases, you can choose this quadrant. But if things are complex, we preferably use this quadrant. And for each quadrant, you can choose which tasks fall in either of these, where you actually need human control in different spectrum, but you are sure that it's better that human is involved.

20:54

Let's say you're going to bake a cake, okay? Technically, human control is needed in different levels. There is not much AI or automation can do for you. Maybe at some part you need oven that takes care of temperature. You need whisker at different parts. But still, this is human who decides. Okay? So depending on the problem, you can choose any of these quadrants. But it's definitely a two-dimensional model, not as previous one, a one-dimensional. So it doesn't mean that if you have high level of automation, you have low level of human control. No, we can have both. And this is what we are after when we talk about human-centered AI. So all of these things lead us to consider ethical AI. You know, ethics is something beyond working well. A system can work well technically. It can have very high accuracy, work very efficiently, give the results very fast. But still, human-centered AI has another aspect which is highly important. This ethical AI means humans should feel they are included in decision-making process. You remember were talking about genders, that people are asked about the genders and the pronoun they want to be addressed by. This is important, or any

problem similar to this should give people a sense of know, spread of categories and people that we are going to include in our AI decision-making process.

23:06

Also, government has lots of regulation and guidelines, and in higher levels, there are some policies have been decided to have a better ethical AI decision. And besides all, new problems keep coming. Okay, so we should keep monitoring AI outputs all the time to see if there is anything that we are missing. Maybe there is a new concern out there these days. You know, by news or things that happen in real world, there might be some new issues that have to be addressed ethically, okay? So all these things lead us to have a better ethical AI output. So that was about human-centered AI. You remember last week I was talking about some modern AI topics. Now I'm going to briefly add more to that about what is these days hot topics in AI, or it's better to say modern AI. One of these topics is responsible AI, which means the AI that we are using should consider many factors and principles to be responsible about what they give us as output. Fairness, privacy, transparency, safety, accountability, and human oversight are the six important principles.

24:30

I will talk briefly about them, but to some extent, we already addressed them a little bit. Fairness simply means the AI shouldn't be biased toward any group of people. And privacy means the data that we collect should be kept safe and secure and do not be used for wrong purposes. Transparency means people should be aware if AI is going to have access to their data. It should be clear to them if you are using AI in your decision-making process. Let's say you're going to a bank and apply for a loan. You should know that if a person is going to read your request or AI is the one who's going to make decision for you. Okay, so there must be transparency here. And by safety, we mean there shouldn't be any harm for a human when you are going to use AI for decision-making process. And accountability, as I mentioned, means it doesn't matter if there is an algorithm and computer who are making a decision. At the very end, you are responsible for this. I believe someone asked this question last week from me. Yes, that's true. People are responsible, and you can never put the blame on algorithm or computer or anything like this. Okay? You are the last layer of decision-making process. That's why we always need human oversight or human intervention in decision-making process.

26:30

So let's talk in a bit more detail. Yeah, as I told you, each of these six principles have some meanings that I decide, but I want you to know that there shouldn't be any discrimination when we talk about fairness, regardless of gender, age, race. OK, everything should be decided by the quality of people, especially when you're going to get a loan, get a job or anything like this. And transparency means people should know if AI is involved in decision-making process. And all these topics I briefly explained. Let's talk about fairness in more detail process. Here I have a good example for you. Assuming there are some people applying for job with different race, different age, and different gender. And even disability, as you see. One of them has disability. Their system only focuses on the quality and merits of the people. If they have the same work experience, same years of experience, same level of expertise, same degree, they should have same score. All of them get 85 out of 100. We assume that people have very similar CVs, resumes. So the gender, the age, and having disability or not shouldn't impact the decision making process. That is a fair AI. OK, as you see, all of them are being viewed equally.

02:49

OK, but there is something important that I want you to focus on. To be fair, your AI should have representative data, which means the data should be collected from all groups equally, which means the number of data points, assuming that you have a table of data, you should have more or less similar number of data points from each gender, each age, and each race. So if you have very few data points regarding people with disability, it really impacts the decision-making process, okay? So the data should be collected equally and fairly in the first place. And you know what? There is a very illustrative example here. Assuming that you have AI and you know that the accuracy is 87.5, that's very high accuracy. Okay. That your AI is... Sorry, can you hear me? Any issue here? I could hear someone, but they stopped talking. So I'll continue. If you have a question, please feel free to write in the chat. We can discuss it in more details after the lecture. It's going to be finished soon, okay? So what I want you to focus here is even having high accuracy doesn't mean that your AI is making a fair decision. For instance, here, by 87.5% accuracy, your model can choose the right person for the job.

30:10

But you know what the problem is? Here, assuming that you have two groups of people, male and female, okay? These two groups have different level of accuracy, okay? For instance, for male, which is technically the majority of data collection, only we have 5% error, but for female, we have 20% error, which means 1 out of 20 is wrong decision for men, and 4 out of 20 is wrong for female. In total, it makes 5 out of 40. 5 out of 40 makes 87.5 accuracy. That's not bad. It's actually good, right? But technically, it's much worse in female case than male. So it's very important that do not simply look at the overall accuracy. We should also have a look at each group or category separately. That's why accuracy is not a very good metric when you have imbalanced data. We have other metrics like F1 score, if you're familiar with that. true negative, true positive, and all these metrics, which gives you a more clear understanding of your AI performance. So it's not only accuracy. Okay? So it's good to bear this in mind when you are dealing with real-world scenarios.

32:06

Okay, let's talk about privacy and data protection. These two are usually used together, but they have a subtle difference. Privacy refers to how private people want to be, okay? They have the right to not give you any personal data. That's why when you are using some AI, they tell you that you're going to be recorded. If you don't like to, please let us know. Okay? So you always have the control and right how far you're going to go by sharing your personal information, okay? That's the meaning of privacy. But with data protection, we mean the data that you collect from people should be kept safe and only used for the purpose that people are aware of. You know, when they are collecting data from people, they can't do that without lots of ethical considerations, okay? They should let people know why they are collecting that data. And after reading all terms and conditions, people decide if they want their data to be shared or not. If they want, they should sign a consent form, which means they agree officially to reveal the data. But still, if they sign the form, it doesn't mean that you can use data anywhere, anytime, for any purpose. You can only use that based on a specific time frame. You can say we keep data for only a year, and after that, we will delete the data, okay? And besides that, you have to specify the data is going to be used for this specific purpose, nothing else. All these things should be very clear in the very beginning. So that's something else that you should be aware. Otherwise, there is very serious consequences for using personal and private data in a wrong way.

34:35

So next things that you should consider is safety, reliability, and misuse of AI when it comes to human-centered AI. Assuming that you design, you know, auto drive car. It doesn't need a driver, but it has some cameras everywhere to look at a street from every different angle to keep it safe. Let's say this AI has 99% accuracy, which is very good, right? But when it comes to human, it means if there is a human obstacle, one out of 100 people can get killed by this car. Right. That's a lot. We shouldn't let such cars drive in the streets. Right. So this is example of safety. We don't want AI to harm people with even 1% chance of happening. Even 0.1% is a lot. So we should be very, very sure that it never happens. By safety, we mean almost nothing, nothing that can harm people. Another example is reliability that we should consider when we are using this for human purposes. The picture that you see is taken from some skin cancer patients. You know, we use biomedical image processing a lot when it comes to detecting skin cancer. But if a model is performing differently in different scenario, you know, same model, same AI, but different performance in different scenario, you cannot rely on that because a reliable AI should work consistently in the area and setting that you specified for this. There is a hospital, patient, same setting, but AI is giving you different outputs. So this is not a reliable one because human safety is above everything.

36:58

And misuse is something else. You know, that AI can perform very well. But what if it's in hands of wrong people? Yep. You know, deepfake is a very good example. Deepfake is performing great for the purpose that it has intended to do, right? It's supposed to create fake videos, and it does that perfectly. But it's not something that we should trust, okay? So as you see, there are many different layers. After having a good system, still there are so many things that you should consider when it comes to human-centered AI, okay? So, next topic was accountability and human oversight. You know, I told you there are many different layers and hierarchy when it comes to human-centered AI. The very first step was what were talking about in all previous weeks. You were supposed to create a model that do some prediction, okay? Or in the very simplest case, predict price of a house, okay, based on location, number of bedrooms, and these criteria or features. That was the very first part that you did as a developer, okay? But still as a developer, you have a lot to be done. You should check if the

model is working well, if you have trained your model with good data, if there is wide variety of, you know, information over there. If there is no bias in the system, all these things should be considered, okay?

39:00

But in the next layer, organization should have a level of authority about everything. They should check the system. They should monitor. They should ensure that there is no failure. They should ensure that if something goes wrong, what will happen. They should write a full documentation about the limitation of the model that you have developed. Okay, all these things should be considered. Then another layer or external companies are involved. They should assess your work. There is another layer when another expert human should check everything. They can approve your decision. If the model is making a very good decision, that's good. They check. Otherwise, if they need some help, they may get back to organization and developers and ask for some more clarification. If they think your model is not doing well, they can override. They can override. They say, no, that is wrong. You should do things differently. Or if they need even more clarification themselves, they can send it to appeal and ask a group of people who decide over that. It's like a jury, okay? Then all these things are one part. In another part, a government and decision makers or policymakers are sitting there. They're also passed some bills how AI should be used. But at the very end, after all these layers, still as a person, this is your responsibility to be very wise when you are using AI. Do not give personal information to your AI, especially your date of birth, or if you have any wrong thought, never share that with AI because, you know, there is a concept we call it profiling. There must be some, let's say, profiling against you. Okay? So be very careful what information you are sharing with AI. After all these layers, still you are responsible for what you are doing with AI.

04:07

So, yeah, here there is a very short activity here for you. I think it's good if you think of an AI who's going to recruit people. It's going to decide who's going to get the job. Okay? Can you please tell me what you think? Just write some keywords in the chat for me. Assuming that there are 1,000 people applying for a job and they need only 20 people for that job. What can go wrong? Assuming that the model has been trained based on previous data, it learned patterns, and it's been trained well on some historical data, then a new person uploads a CV and AI is going to give a score to them.

And they decide if this person is a good choice or not. If person is a good choice, there is another layer of interview. They are shortlisted, then there is human interview. But even at this level, AI can exclude some people. Please tell me how can ensure this AI is doing a good job or what can be improved? I give you a couple of minutes. Think about that and type some keywords in the chat. Yes, Kathleen, I definitely agree with you. There are many examples of, you know, misusing AI intentionally or unintentionally. Sometimes people have very good will. They want to do the right thing, but this is AI who's making bad decisions.

43:13

So if you look at the example that I made here for you, the activity, as a person who is aware, you know much more than most people in this street about AI, right? After today, you will know even better and more. What do you think can improve these issues? Because some companies, they really have a very good will. They want to do things right, but it's not possible. How we can get rid of these issues? How we can have more trust for the AI? Yes, that's true about Amazon, Kathleen, you're right. Yes, Alex, that's perfectly correct. Yep. You remember I was talking about representative data. Opposite that, we have under or even unrepresented data. Under means you have less number of case. Let's say we have much fewer examples of darker skin people, especially female one. This is under-representative data, but un-representative data is even worse. That means there is not any case of them. So that's very important, how data has been prepared, filtered, cleaned. That's what we discussed in previous weeks. You should have balanced number of data points from each category to have a fair decision. So Kathleen says the AI should also be reviewed before it's implemented. Yes, that's true. Yes, absolutely true. Alexa, something else? Yeah, human oversight. Yeah, yes.

45:14

That's all true, guys. So, you know, that is a very nice area if you want to work on. There is a course at Deakin, human-centered AI. They may not have the exact name for this, but they have it in the title. I think it's good and interesting if you want to know more about these topics there. So all these things are right, guys, and we are developing, we are progressing towards having a human-centered AI, which means everything is for human, okay? AI is there to empower us, to let us have a better experience, not getting our jobs or impacting us or deceiving us, okay? So that's the whole purp

use of human-centered AI. Something else here. Could you also make it so the AI only sees information about experience and qualification? It's blind. That is possible, but you know, sometimes it's not that simple, Alex, you know. Let's say they are going to make a decision about what person they should hire. Okay? They may not ask about their socioeconomic situation, which means which social status they have or what background they have, if they're coming from wealthy families or poor families. Technically, they don't ask these questions. But sometimes they get the data incorrect, sorry, let's say indirectly, okay, indirectly. And they may not be even obvious to them. But when they ask for address based on the zip code, AI can automatically label people. They can impact the decision-making process. So it's not that simple if they don't ask about gender. It doesn't mean anything. But the data is there. Even if I make data anonymous, still the data is there. Okay? So it's much more important than this.

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That's why, despite AI is progressing well, still I think there is a lot that human can do to make it more trustworthy. So never think that AI is there, it's done, and we can do anything. Still, there is a lot that should be done. There is a huge room for improvement. That's why I hope you future data scientists can do a great job. Yeah, my pleasure, Alex. My pleasure. So this is much more complex than what it seems, you know. So let me go to the slides. Yeah, thank you for the activity. So the last part of this unit, guys. This AI agent is one of the most hot topics that we are dealing with AI. AI agent is probably something that is moving toward independent AI, where AI can do things by itself. Okay? I'm going to talk about this, how it works, and give you some examples. And at the end, I tell you how this AI agent is different from AI models that we talked about so far.

49:02

So what is AI agent? AI agent is a system, okay? So it's not an isolated part. AI model is just a piece of code, okay? But AI agent is something much more comprehensive than this, okay? It's really sophisticated and it can make decision and also work towards that decision. OK, so human gives it a goal. It has, I'll explain in next slides, but simply know that it has a goal. it's interacting with one environment under a specific setting, and it makes decisions and takes action, okay? And it observes the result of its action, takes feedback, and based on the feedback, it tries to improve its action. So it's a circle of work, okay? It observes, makes some reasoning, acts accordingly, learn

s from the feedback, and then makes another decision to improve previous action. So you see, it's exactly like what a human does, right? Or when a baby starts to learn how to walk, they all go through this process. That's exactly what agent does. OK, so a cycle can repeat until goal is achieved or until a condition has met where human programmed or asked agent to stop if you reach or see something like this, stop it and call for human help. Okay, that's how human and automation can be merged together. Despite agent is highly automated, still human can be highly involved. So that's probably the best combination of human and AI that we have so far.

05:07

So it's worth to know more about that. That's the cycle that agent is performing. First, a human can give it a definite objective or goal, okay? I have example for you that I explain more clearly. So that's why I'm skipping the gray highlighted part at this stage. Human gives a goal to agent, okay? Then the agent observes this goal and environment that it's facing. It collects data from environment, which means observation. It sees what is happening in this setting or environment, and the other eye is on the goal. I have to achieve this from what I already have. Okay? So here and there is observed by agent. Okay? I know that as a human, I know that I have this weight. My aimed weight is like to lose 10 kilos. Okay? So I have a goal and I have some observation of my current situation. Then I plan. Agent does exactly this. It plans towards the goal. Okay, and it decides what action can be taken to achieve that goal. Then it acts accordingly based on the action that it decided. It acts something, okay? And then it observes what happened. It evaluates itself by getting feedback from environment. That's exactly what I do. I change one specific, I choose one diet. After a month, I see what was the impact of diet on my weight. If my weight has not changed much, probably I decide otherwise. I choose another diet, right? That's exactly what AI or agent does, okay? So it's mimicking human behavior here.

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So after collecting feedback and evaluation, it again observes what happened. Change the plan and act differently until it achieves the goal. Or if something goes wrong, it can stop the process and call for human intervention. You see, this is almost including everything that we wanted so far from AI. So AI is highly practical in many different aspects. Many different aspects. Even in farming or agriculture, AI or AI agents are highly decisive these days. So let's have a closer look at the components. You know, t

he goal is being fed to AI or agent, as you see. This is like, this is not a robot, actually. This is different from robots. A robot does physical things. For instance, assuming that I have a robot that cooks for me. It does physical things. But agent is different. It thinks smartly. So by environment here, we simply mean the setting. Let's say I'm going to ask agent to schedule a meeting for me and my PhD students. The environment will be the calendar of all of us. It goes to my email and understands my PhD students. Then it checks the calendar of all of us and tries to find a vacant time slot that suits us the best. Okay? Then it creates a meeting, something like this. And all these things are happening here.

55:24

So, as you see, I think this slide is highly repetitive. I explained all sections. Only the memory is something new probably here. Memory is like human memory. It needs to learn from previous actions. I told you that it collects feedback from environments, okay? So it needs a memory to save these learning processes there. And it's always there because it's running on a computer, okay? But it's good to know that's how AI agent works, and all these components are involved. And as we expected, at the very end, we have human intervention that makes things the best. So talking about the example that I already gave you, let's think about one practical example. When I ask AI agent to schedule a meeting among me and my own PhD students. So first things it does is reading my email to see who is the PhD student I'm dealing with. So I don't even have to give the names. It check and realized who is or are involved. Okay, after identifying the participant, it checks all of us calendars to find a suitable or vacant time slot, then propose a time, and so far it has taken all of several steps. Then here, it tries to get feedback from me by evaluating what is my response, if I approve it or not. If I don't, it means evaluation gives a feedback that says it's wrong. Then it repeats the process. It tries to choose another time slot until I say it's good. Okay? If their human intervention says stop it, that's nice. Then it can take action towards its goal and create a meeting and send invitation to everyone.

57:54

So that's how agents can be helpful and make things more efficient while having human in the center of decision-making process. So AI does the hard job. And it's up to me to approve it or not. And that's the best approach that we can expect from agents. But there is a very delicate difference here between AI model and AI agent. You know

ow that AI model is like a piece of code that you write, okay? It simply takes input and gives you output. Just remember the example that I made for predicting house price. You give a data set, you ask a question, it gives you a predicted price for a house. Input, output process. But AI agent is much more than that. As you noticed, it works towards a goal, okay? And it can perform action. It can make decision. AI simply makes a decision and it's up to us to take it or not. But agent takes action according to what they decide, okay? And they can repeat the process several times toward the goal. They also can use APIs and some databases to evaluate the performance. So that's much more comprehensive than what AI does, okay, a simple AI model does, okay?

59:28

And the very good example is here. Even generative AI like Gemini, ChatGPT, all these models that were talking previously, despite being highly complex and sophisticated, at best, regarding writing email, for instance, they can simply summarize an email for you. You cannot tell ChatGPT that, oh, write email to my PhD students. Okay? You can't do that. But AI agent can do that. You can ask it, read my email and answer them, tell them that there is no meeting today. That's it. With one single sentence, an AI agent can understand what the goal is. They can collect information that it needs itself. It can make a decision. If you approve, it will write email and send email to everyone. So you see, this is much more sophisticated and comprehensive. So that's the best things we have so far. But still, it has some issues. There are some risks. As you could guess, probably one of the worst ones is incorrect or unauthorized action. When wrong people are using good tools, like, you know, a knife. If a surgeon is using this, it can save people's lives. But what if someone who has mental issues is using this? It can harm himself or others, right? So it's a double-edged sword. We always should be careful who is using AI agents.

06:07

Again, same issues with personal data. People should be very careful about sharing their information with AI agents. It may go through a vicious circle, continuously give wrong answers. Let's say I'm asking to create a meeting, but it keeps coming to me, okay, this time a slot, and I say, no, that's not good. And I say, no, that's not good. It tries to find another one, which is not good either. It tries to find another one, which is not good either. So what if it gives me not appropriate time? Or what if there is not any appropriate time at all? So it's also a bad scenario that can happen. And this do

mino effect is very important. Sometimes small mistakes can compound and accumulate toward a very serious bad decision. Okay? Because this is not human anyhow. Still, we are not there yet to consider or include all safety, responsibility, accountability, transparency, and all these concerns that we have to have a responsible AI, okay? So there are many concerns and risks that we should include. One of them is ambiguous goal. Let's say I'm telling something that has a little bit of sarcastic tone in that. AI is not able to understand. If I make a joke, for instance, it cannot understand, and it may choose a wrong goal, okay? If I say something clear and formal, it gets the goal. But if not, it can set a wrong goal, okay? So there are many risks that is involved in using AI agents.

01:03:35

Even being the best things we have so far, still we are not there. That's why last week when we were talking, I told you that don't worry, AI is not there yet to steal your jobs or your future or anything like this. I think technically, as further we go, there is more we can do with AI as human. So I invite you to learn more, understand more, and work towards to be creative data scientists. Okay? And for you as younger generation, I hope you come up with better solution for AI. So guys, thank you so much for your attendance. That was the last lecture we had in this unit. I hope you enjoyed, and I love to hear your feedback. Probably you received email for evaluation. I love to hear what you think of my lecture, the way that I was teaching. If you have any recommendation, I love to hear that. So I'm happy to answer your questions. I probably better to stop sharing and stop recording first. Then we can talk about your concern, your questions, and if you have any questions regarding portfolio.